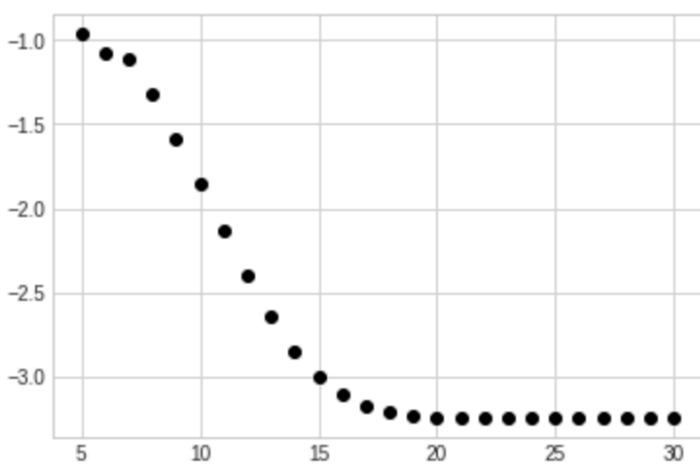
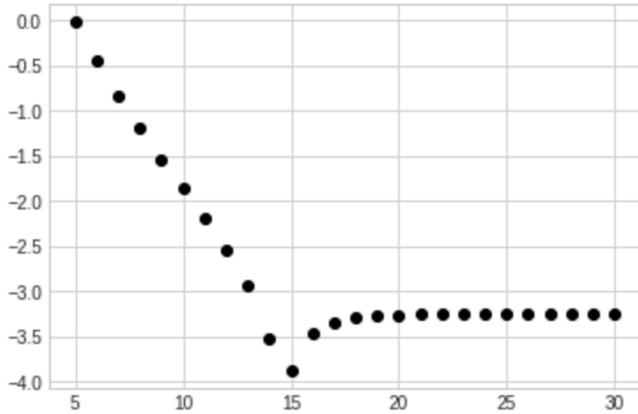
[3]:

```python
%matplotlib inline
import matplotlib.pyplot as plt
plt.style.use('seaborn-whitegrid')
import numpy as np
import math
import sys
def f(x):
    return math.sin(4.8*math.pi*x)
def forwardDiff(value,k):
    return (f(value+k)-f(value))/k
def backwardDiff(value,k):
    return (f(value)-f(value-k))/k
def centralDiff(value,k):
    return (f(value+k)-f(value-k))/(2*k)
def customDiff(value,k):
    return (2*f(value + k) + 3*f(value) - 6*f(value-k) + f(value-2*k))/(6*k)
fd=0
bd=0
cd=0
custom=0
#trueVal computed by hand
trueVal=-14.96017024
errorsF=[]
errorsB=[]
errorsC=[]
errorsCustom=[]
ks=[]
for k in range(5,31):
    ks.append(k)
    fd=forwardDiff(0.2,2 ** -k)
    bd=backwardDiff(0.2,2 ** -k)
    cd=centralDiff(0.2,2 ** -k)
    custom=customDiff(0.2,2 ** -k)
    errorsF.append(abs(fd-trueVal))
    errorsB.append(abs(bd-trueVal))
    errorsC.append(abs(cd-trueVal))
    errorsCustom.append(abs(custom-trueVal))
```

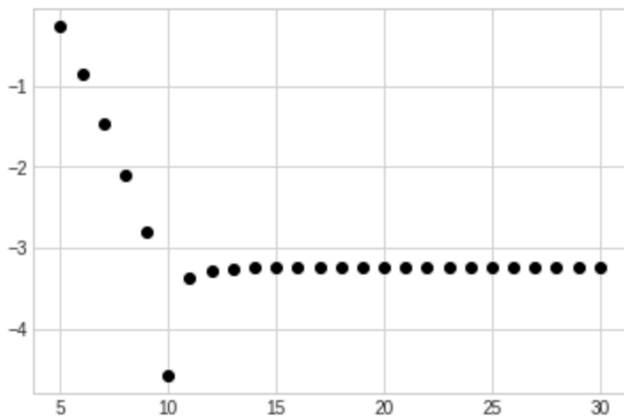
```python
logErrorsF=[]
logErrorsB=[]
logErrorsC=[]
logErrorsCustom=[]
for i in range(0,26):
    logErrorsF.append(math.log(errorsF[i],10))
    logErrorsB.append(math.log(errorsB[i],10))
    logErrorsC.append(math.log(errorsC[i],10))
    logErrorsCustom.append(math.log(errorsCustom[i],10))
plt.plot(ks, logErrorsF, 'o', color='black');
```



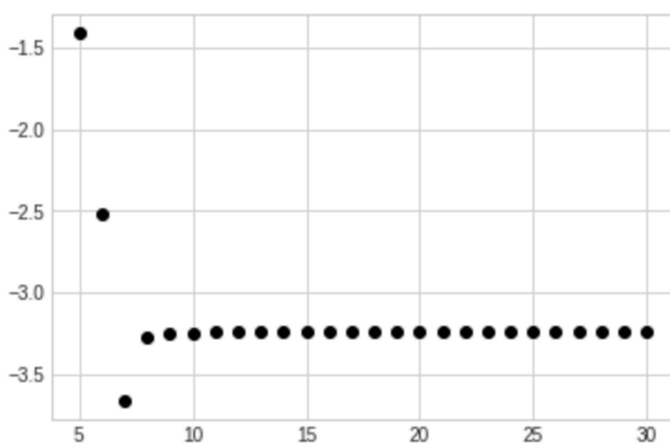
[4]:
```
plt.plot(ks, logErrorsB, 'o', color='black');
```



[5]:
```
plt.plot(ks, logErrorsC, 'o', color='black');
```



[6]:
```
plt.plot(ks, logErrorsCustom, 'o', color='black');
```



K values of asymptotic regimes:

Forward Approximation: 7-14
Backward Approximation: 5-13
Central Approximation: 5-9
Custom Approximation: 5-7

The Central approximation appears to converge twice as fast as the forward and backward approximations. The custom approximation appears to converge twice as fast as the central approximation.